Supplementary material

1 Batch size parameter

Self-supervised loss functions (e.g. NT-Xent and CLIP loss function) are sensitive to batch size adopted during the model training. The performance of the multimodal architecture is tested on the classification and retrieval task. In both tasks, larger batch sizes lower the performance of the multimodal architecture. Table 1 shows an overview of the classification performance. On this task, the highest performance is reached adopting a batch size equal to 4. The reason may be identified in the limited amount of samples used for training the architecture: small batch sizes increase the iterations of the training algorithm and therefore the network weight updates.

A similar scenario can be identified considering the multimodal retrieval task, considering both the retrieval of reports and WSIs. In the first case (retrieval of reports using WSIs as input), the application of different batch sizes adopted during training shows multiple effects, depending on the setup used to train the multimodal architecture. When the architecture is trained using only the self-supervised CLIP loss function, to align image and report representations, larger batch sizes lead to slightly higher performance, in terms of precision@k. This result shows how the CLIP loss function benefits of large batch sizes. On the other hand, the architecture trained only optimizing the self-supervised loss function proposed in the paper shows the opposite behavior: larger batch sizes lead to poorer performance. When the architecture is trained to optimize a classification term, the effect of increasing the batch size is similar to what has been shown on the classification performance: small batch sizes (4-8) lead to higher performance, compared with the one obtained adopting larger (16-32) batch sizes. In the second retrieval case (retrieval of WSIs using reports as input) the performance follows the pattern identified in other retrieval cases: the highest performance is reached adopting a batch size equal to 4. When the architecture is trained using only the self-supervised CLIP loss function, the performance is comparable and does not statistically significant improve varying the batch size. On the other hand, the architecture trained only optimizing the self-supervised loss function dramatically decreases the retrieval performance, when larger batch sizes are adopted. When the architecture is trained to optimize a classification term, the effect of increasing the batch size slightly modifies the performance, but not dramatically.

Table 1: Overview of the results on the classification performance, adopting different batch-size parameters during the architecture training. The comparison involves three training setups: unimodal (training only with WSIs), multimodal (trained using CLIP loss to align image and report representations) and multimodal (trained using our loss function to align image and report representations). The performance is evaluated with weighted F1-score, reporting the average and the standard deviation (of the models involved in the k-fold cross-validation).

	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Unimodal	0.765 ± 0.013	0.771 ± 0.012	0.757 ± 0.031	0.772 ± 0.015	0.760 ± 0.011	0.763 ± 0.014	0.750 ± 0.018	0.744 ± 0.023
Multimodal (CLIP)	0.785 ± 0.013	0.782 ± 0.013	0.768 ± 0.011	0.762 ± 0.017	0.765 ± 0.014	0.755 ± 0.013	0.744 ± 0.025	0.731 ± 0.021
Multimodal (our)	0.788 ± 0.011	0.792 ± 0.012	0.774 ± 0.018	0.780 ± 0.013	0.782 ± 0.011	0.785 ± 0.009	0.769 ± 0.012	0.772 ± 0.010

Table 2: Overview of the results on the multimodal retrieval performance, adopting different batch-size parameters during the architecture training. The task involves the retrieval of reports, using a WSI as input. The comparison involves four training setups: self-supervised (trained using CLIP loss to align image and report representations), self-supervised (trained using our loss function to align image and report representations), multimodal (trained using CLIP loss to align image and report representations) and multimodal (trained using our loss function to align image and report representations). The performance is evaluated with precision@k (k values are 5, 10, 50) and Mean Average Precision (k is 1000), reporting the average and the standard deviation (of the models involved in the k-fold cross-validation).

Retrieve reports (images as input)								
Self-supervised (CLIP)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.255 ± 0.030	0.244 ± 0.097	0.287 ± 0.039	0.267 ± 0.057	0.285 ± 0.034	0.232 ± 0.072	0.292 ± 0.071	0.275 ± 0.098
Precision@10	0.253 ± 0.031	0.251 ± 0.076	0.290 ± 0.029	0.274 ± 0.063	0.276 ± 0.036	0.228 ± 0.06	0.290 ± 0.07	0.267 ± 0.095
Precision@50	0.249 ± 0.049	0.268 ± 0.064	0.283 ± 0.038	0.280 ± 0.051	0.261 ± 0.022	0.234 ± 0.048	0.272 ± 0.033	0.261 ± 0.058
mAP	0.179 ± 0.024	0.212 ± 0.040	0.190 ± 0.027	0.219 ± 0.036	0.177 ± 0.015	0.199 ± 0.023	0.171 ± 0.017	0.204 ± 0.018
Self-supervised (our)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.757 ± 0.021	0.768 ± 0.020	0.542 ± 0.103	0.549 ± 0.104	0.507 ± 0.056	0.507 ± 0.057	0.526 ± 0.057	0.485 ± 0.037
Precision@10	0.756 ± 0.019	0.768 ± 0.021	0.554 ± 0.096	0.549 ± 0.098	0.512 ± 0.050	0.511 ± 0.05	0.532 ± 0.051	0.485 ± 0.034
Precision@50	0.770 ± 0.009	0.775 ± 0.016	0.58 ± 0.087	0.559 ± 0.09	0.526 ± 0.038	0.515 ± 0.041	0.538 ± 0.044	0.485 ± 0.025
mAP	0.463 ± 0.017	0.583 ± 0.008	0.367 ± 0.046	0.428 ± 0.066	0.349 ± 0.017	0.397 ± 0.028	0.340 ± 0.014	0.365 ± 0.025
Multimodal (CLIP)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.780 ± 0.012	0.777 ± 0.016	0.772 ± 0.029	0.758 ± 0.021	0.779 ± 0.008	0.765 ± 0.02	0.784 ± 0.012	0.762 ± 0.019
Precision@10	0.779 ± 0.014	0.779 ± 0.018	0.773 ± 0.028	0.761 ± 0.02	0.775 ± 0.007	0.765 ± 0.018	0.783 ± 0.013	0.763 ± 0.017
Precision@50	0.789 ± 0.010	0.789 ± 0.017	0.776 ± 0.018	0.778 ± 0.014	0.778 ± 0.007	0.774 ± 0.016	0.782 ± 0.012	0.769 ± 0.018
mAP	0.546 ± 0.020	0.639 ± 0.018	0.512 ± 0.015	0.609 ± 0.011	0.494 ± 0.021	0.604 ± 0.015	0.489 ± 0.019	0.601 ± 0.014
Multimodal (our)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.795 ± 0.030	0.794 ± 0.015	0.785 ± 0.029	0.788 ± 0.012	0.796 ± 0.020	0.792 ± 0.013	0.793 ± 0.010	0.779 ± 0.015
Precision@10	0.794 ± 0.028	0.796 ± 0.015	0.785 ± 0.028	0.790 ± 0.011	0.795 ± 0.019	0.791 ± 0.013	0.791 ± 0.011	0.780 ± 0.016
Precision@50	0.798 ± 0.020	0.802 ± 0.013	0.791 ± 0.022	0.791 ± 0.012	0.792 ± 0.018	0.791 ± 0.009	0.788 ± 0.008	0.781 ± 0.012
mAP	0.537 ± 0.023	0.632 ± 0.012	0.516 ± 0.018	0.618 ± 0.014	0.505 ± 0.017	0.611 ± 0.012	0.502 ± 0.007	0.601 ± 0.009

Table 3: Overview of the results on the multimodal retrieval performance, considering different batch-size parameters during the architecture training. The task involves the retrieval of WSIs, using a report as input. The comparison involves four training setup: self-supervised (trained using CLIP loss to align image and report representations), self-supervised (trained using our loss function to align image and report representations), multimodal (trained using CLIP loss to align image and report representations) and multimodal (trained using our loss function to align image and report representations). The performance is evaluated with precision@k (k values are 5, 10, 50) and mAP, reporting the average and the standard deviation (of the models involved in the k-fold cross-validation).

Retrieve images (reports as input)								
Self-supervised (CLIP)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.233 ± 0.066	0.231 ± 0.059	0.237 ± 0.062	0.257 ± 0.046	0.264 ± 0.084	0.243 ± 0.048	0.236 ± 0.047	0.260 ± 0.058
Precision@10	0.248 ± 0.049	0.239 ± 0.041	0.218 ± 0.053	0.257 ± 0.032	0.251 ± 0.062	0.240 ± 0.036	0.245 ± 0.057	0.261 ± 0.054
Precision@50	0.247 ± 0.031	0.245 ± 0.026	0.220 ± 0.029	0.267 ± 0.012	0.243 ± 0.045	0.251 ± 0.016	0.248 ± 0.045	0.262 ± 0.026
mAP	0.159 ± 0.008	0.186 ± 0.007	0.16 ± 0.009	0.190 ± 0.012	0.164 ± 0.015	0.186 ± 0.015	0.163 ± 0.014	0.189 ± 0.015
Self-supervised (our)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.836 ± 0.030	0.850 ± 0.016	0.779 ± 0.045	0.801 ± 0.034	0.754 ± 0.042	0.744 ± 0.042	0.685 ± 0.074	0.614 ± 0.092
Precision@10	0.832 ± 0.028	0.851 ± 0.012	0.771 ± 0.046	0.800 ± 0.029	0.751 ± 0.032	0.740 ± 0.037	0.679 ± 0.066	0.608 ± 0.079
Precision@50	0.827 ± 0.012	0.845 ± 0.007	0.747 ± 0.034	0.775 ± 0.033	0.73 ± 0.024	0.712 ± 0.037	0.663 ± 0.05	0.591 ± 0.062
mAP	0.467 ± 0.010	0.580 ± 0.010	0.357 ± 0.041	0.433 ± 0.050	0.323 ± 0.016	0.371 ± 0.029	0.292 ± 0.024	0.318 ± 0.022
Multimodal (CLIP)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.847 ± 0.033	0.887 ± 0.016	0.836 ± 0.017	0.872 ± 0.016	0.85 ± 0.009	0.877 ± 0.008	0.839 ± 0.028	0.870 ± 0.012
Precision@10	0.847 ± 0.024	0.882 ± 0.017	0.833 ± 0.016	0.867 ± 0.012	0.847 ± 0.010	0.874 ± 0.007	0.835 ± 0.021	0.870 ± 0.01
Precision@50	0.846 ± 0.013	0.874 ± 0.014	0.830 ± 0.011	0.862 ± 0.013	0.840 ± 0.011	0.864 ± 0.009	0.828 ± 0.013	0.855 ± 0.007
mAP	0.515 ± 0.020	0.632 ± 0.022	0.480 ± 0.016	0.596 ± 0.018	0.489 ± 0.011	0.586 ± 0.015	0.483 ± 0.008	0.571 ± 0.008
Multimodal (our)								
	4		**8**		**16**		**32**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Precision@5	0.867 ± 0.030	0.880 ± 0.015	0.850 ± 0.017	0.892 ± 0.009	0.847 ± 0.022	0.869 ± 0.010	0.846 ± 0.014	0.870 ± 0.010
Precision@10	0.864 ± 0.024	0.877 ± 0.013	0.847 ± 0.016	0.886 ± 0.011	0.846 ± 0.017	0.868 ± 0.012	0.842 ± 0.012	0.866 ± 0.009
Precision@50	0.854 ± 0.021	0.873 ± 0.016	0.848 ± 0.011	0.874 ± 0.009	0.834 ± 0.014	0.865 ± 0.009	0.836 ± 0.011	0.858 ± 0.007
mAP	0.508 ± 0.019	0.620 ± 0.015	0.508 ± 0.019	0.617 ± 0.016	0.506 ± 0.010	0.605 ± 0.010	0.501 ± 0.007	0.603 ± 0.008

2 Training the architecture with unpaired WSIs/reports

The multimodal architecture shows an increase in WSI classification performance even in the scenario where not all the available WSIs and reports used to train the model are paired.

The collection of WSIs and pathology reports paired together may not be trivial, especially in the scenario where data are collected from multiple sources, which do not provide both modalities. The multimodal architecture is evaluated with the weighted F1-score, considering different percentages of paired WSI-report input data and reporting the average and the standard deviation of ten models (trained using cross-validation). The dataset adopted includes only WSI-report samples, paired together. In order to simulate a scenario where some samples are not paired together, a percentage of training data is used only to train the model on the classification of images or texts, without the modality alignment terms. The architecture is trained with an increasing percentage of data, starting from 20% and up to the whole dataset. The training results, on the increasing amount of data, are shown in Table 4.

Table 4: Results for the performance of the multimodal architecture trained with an increasing amount of paired WSI-reports on the classification of WSIs. The evaluation involves the Catania cohort, Radboudumc datasets and their combination (cumulative). The multimodal architecture is trained optimizing the classification terms and our self-supervised loss function. The performance is evaluated with weighted F1-score, reporting the average and the standard deviation (of the models involved in the k-fold cross-validation).

Percentage paired samples	**Catania**	**Radboudumc**	**Cumulative**
0% (unimodal)	0.765 ± 0.013	0.771 ± 0.012	0.769 ± 0.011
20%	0.776 ± 0.009	0.782 ± 0.010	0.780 ± 0.008
40%	0.782 ± 0.008	0.784 ± 0.018	0.784 ± 0.009
60%	0.767 ± 0.016	0.785 ± 0.012	0.778 ± 0.012
80%	0.783 ± 0.008	0.787 ± 0.015	0.785 ± 0.012
100% (multimodal (our))	0.788 ± 0.011	0.792 ± 0.012	0.790 ± 0.009

The multimodal architecture trained with percentages of paired WSI-report samples shows higher performance, compared with the same architecture trained only on WSIs, starting from 20% of paired data. These results may be the consequence of two aspects. The first one is that the architecture slightly benefits of two modalities, even when the self-supervised loss functions are not systematically applied to align multimodal data representations (lower percentages), since the architecture includes some layers shared from both modalities. The second one involves the effect of multimodal sample alignment: when the whole dataset is aligned, the classification performance is higher than in the other cases. Also in this case, the shared layers may be the consequence of this result: when WSIs and reports are aligned, they show similar embedding vectors, which are the input for the last fully connected layers, including the classifier.

3 Linking between visual and textual concepts

A detailed overview of the linking between visual and textual concepts, considering only the multimodal setup, trained with our loss function (L1-loss + cosine similarity loss + NT-Xent loss) to align image and report representations.

The linking is effective, considering the global accuracy and the fact that the concepts were not used during the training. In particular, classes such as 'Tubular Adenoma', 'Tubolovillous Adenoma', 'Pre-Cancerous Dysplasia', 'Mild Dysplasia', 'Severe Dysplasia', reached high-level accuracy. On the contrary, other classes reached a low-level, such as 'Polyp'. This scenario can be explained considering the region included in a single patch, analyzed by the architecture: the patches are small in size (224x224 pixels, extracted from magnification 10x), therefore tissue structures such as polyps can be entirely captured.

Table 5: Overview of the results on the linking between visual and textual concepts, considering the single concepts. The setup adopted is the multimodal one, trained with our loss function to align images and reports. For every concept, one hundred images are collected and reviewed by an expert pathologist. The model adopted for the linking is the one reaching the highest performance in terms of classification, among the training repetitions involved in the k-fold cross-validation.

Concept	**True Positives**	**Accuracy**
Adenocarcinoma	63/100	0.63
Adenocarcinoma (Invasive adenocarcinoma)	63/100	0.63
Adenoma	86/100	0.86
Dysplasia	61/100	0.61
High Grade Dysplasia	64/100	0.64
Hyperplastic Polyp	21/100	0.21
Mild Dysplasia	81/100	0.81
Moderate Dysplasia	99/100	0.99
Polyp	0/100	0.0
Pre-Cancerous Dysplasia	73/100	0.73
Serrated Adenoma	11/100	0.11
Severe Dysplasia	85/100	0.85
Tubular Adenoma	99/100	0.99
Tubulovillous Adenoma	100/100	0.100
Villous Adenoma	5/100	0.05
Total	**911/1500**	**0.607**

4 Classification as zero-shot learning task

The architecture can be adopted as a WSI classifier even when it is trained without a classifier, such as in the self-supervised setups. The classification is tackled as a zero-shot learning: for each class, an embedding vector representing the textual concept is compared with the embedding representing a WSI, using the cosine similarity. The classes are linked to the images when the similarity is high. Since the WSIs are labeled with multimodal annotations, several similarity thresholds (0.7, 0.8, 0.9) are tested: a concept is linked to a WSI only when its similarity exceeds the threshold.

Table 6 shows an overview of the classification performance when zero-shot learning is applied. The multimodal architecture, trained optimizing a classification term, reaches the highest result. This result is not surprising, since the model is explicitly trained to classify images and reports. In this case, the highest results are reached when the threshold is equal to 0.7 and 0.8. Particularly, in the first case, the highest performance is reached by the multimodal setup, adopting our loss function. The case where the architecture is trained using only self-supervised loss functions to align image and report representations is more interesting. When the alignment is achieved via CLIP loss optimization, the performance reached is poor. This is a consequence of the fact that the similarity between classes and images is always too low to exceed the threshold and it shows how CLIP loss function can be hardly adopted when training datasets are not large in size. On the other hand, when the architecture is trained only using our loss function to align modalities, the result is higher. The highest performance is reached with a threshold equal to 0.7: on Catania data, the model reaches a F1-score equal to 0.598 ± 0.082, while instead on Radboudumc data, the model reaches a F1-score equal to 0.690 ± 0.029. These results must be highlighted, considering the absence of classification in the model training and the relative gap with the classification performance of the architecture, trained only on WSI classification (respectively F1-score equal to 0.765 ± 0.013 on Catania and 0.771 ± 0.012 on Radboudumc).

Table 6: Overview of the results on the WSI classification, via zero-shot learning, considering all multimodal setups. The performance is evaluated on the test partition collected from pathology workflow (Catania and Radboudumc). Being a multilabel problem, three threshold levels for the matching are proposed: 0.7, 0.8, 0.9. The performance is evaluated with weighted F1-score, reporting the average and the standard deviation (of the models involved in the k-fold cross-validation).

	0.70		**0.80**		**0.90**	
Setup	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**	**Catania**	**Radboudumc**
Self-supervised (CLIP)	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
Self-supervised (our)	0.598 ± 0.082	0.690 ± 0.029	0.406 ± 0.105	0.674 ± 0.076	0.164 ± 0.058	0.357 ± 0.124
Multimodal (CLIP)	0.615 ± 0.029	0.622 ± 0.038	0.592 ± 0.169	0.702 ± 0.071	0.479 ± 0.198	0.523 ± 0.073
Multimodal (our)	0.715 ± 0.022	0.702 ± 0.027	0.628 ± 0.091	0.759 ± 0.024	0.262 ± 0.066	0.549 ± 0.094